

Cyclistic Bike-share Case Study

Corina Mocanu
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Objective

The purpose of this analysis is to understand how annual members and casual riders use Cyclistic bikes differently in order to support the development of marketing strategies that could convert casual riders into annual members.

The project analyzes historical Cyclistic trip data covering previous 12 months, from May 2025 to April 2026. The data was downloaded from Motivate International Inc, on May 24, 2026.

Original dataset: 5.7+ million rides

Final dataset: 5 541 674 rides

Tools

- Microsoft Excel
- Power Query
- Excel Data Model
- Pivot Tables
- Pivot Charts
- Microsoft PowerPoint

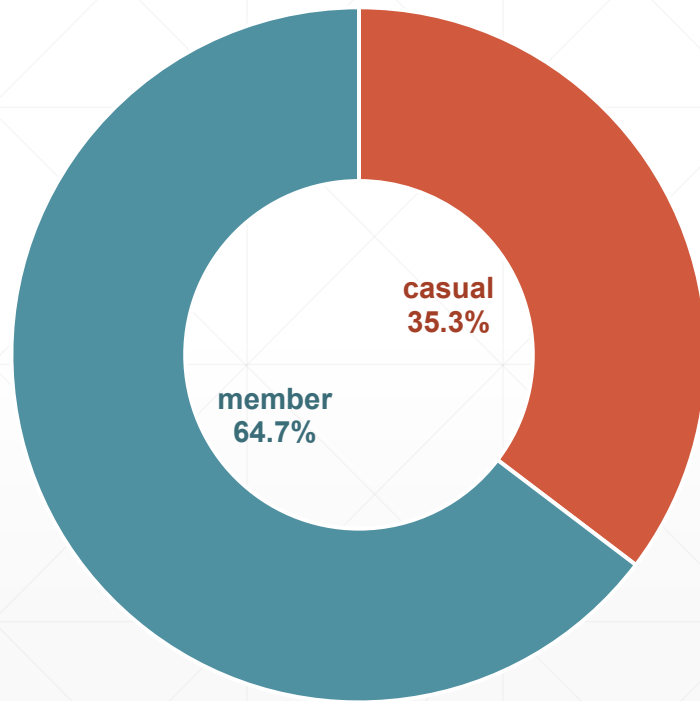
Methods

- Data cleaning and validation
 - Duplicate detection
 - Data transformation
 - Feature engineering (ride_length, day_of_week)
 - Exploratory Data Analysis (EDA)
 - Aggregation and trend analysis
 - Data visualization
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Data Cleaning Summary

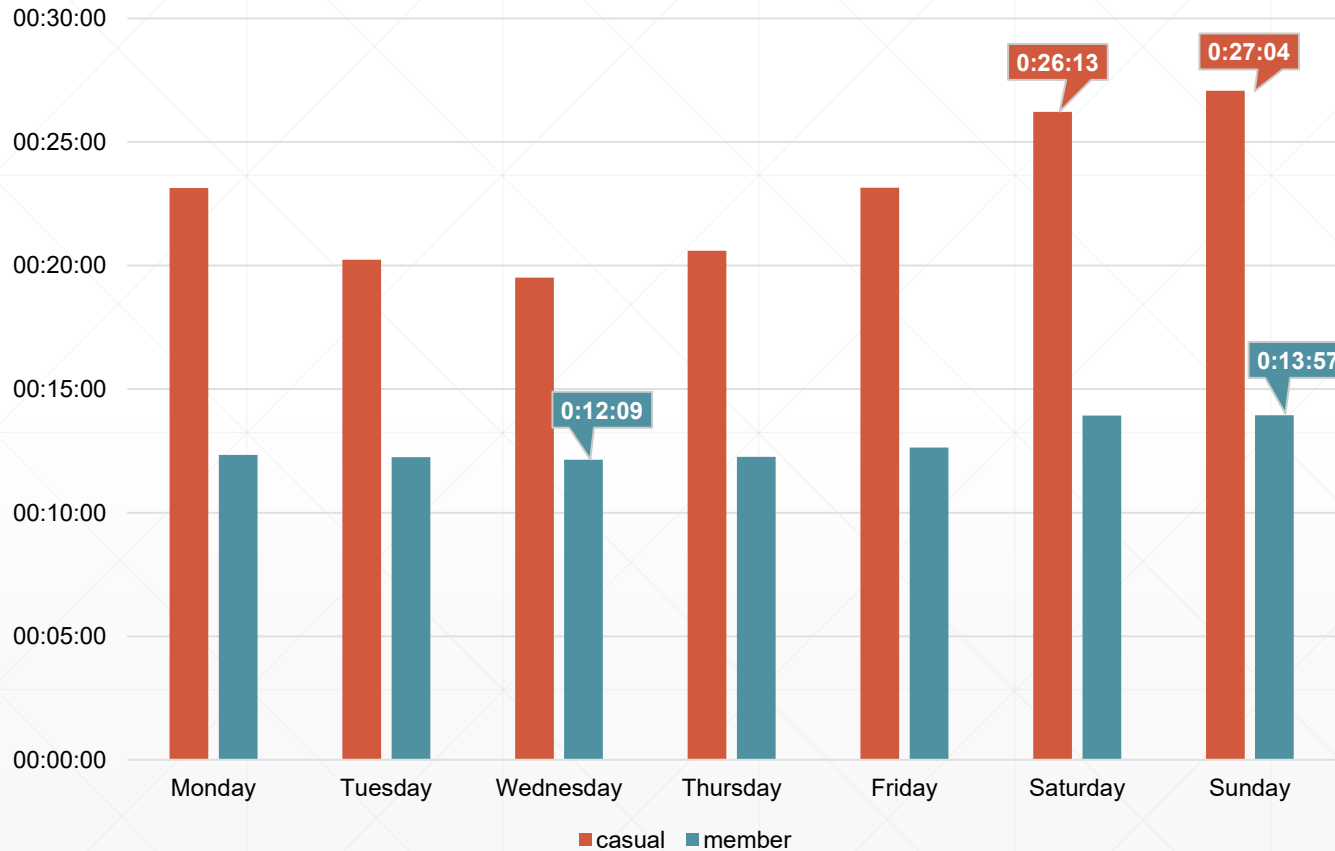
- Excluded station and coordinate fields due to missing values
 - Standardized datetime formats
 - Removed duplicate records
 - Created ride_length variable
 - Created day_of_week variable
 - Removed 155 781 rides shorter than one minute to reduce the impact of invalid or test rides
 - Combined 12 monthly datasets
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Ride share by customer type



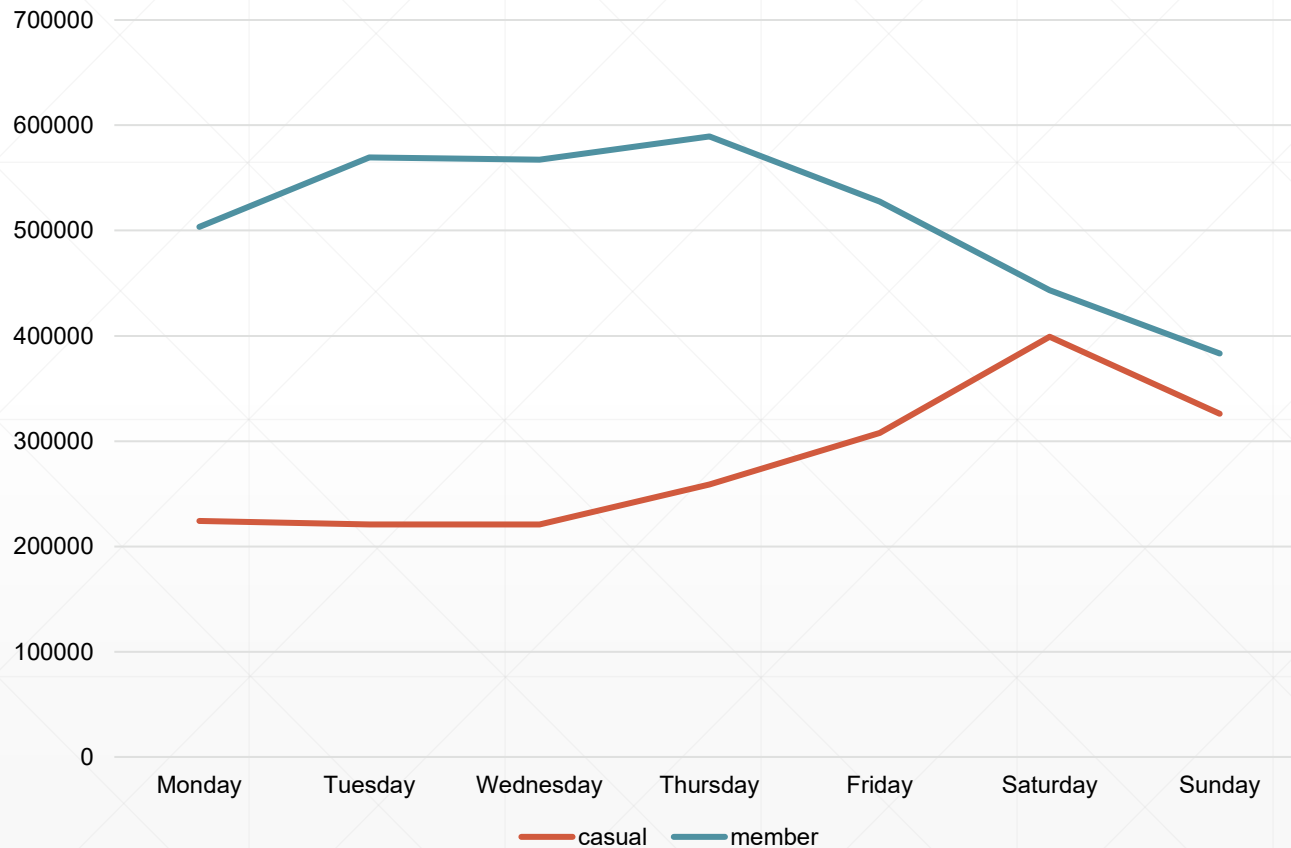
- Annual members account for 64.7% of all rides.
 - Casual riders account for 35.3% of all rides.
 - Despite representing a smaller share of rides, casual riders spend significantly more time per trip than annual members.
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Average ride length by weekday



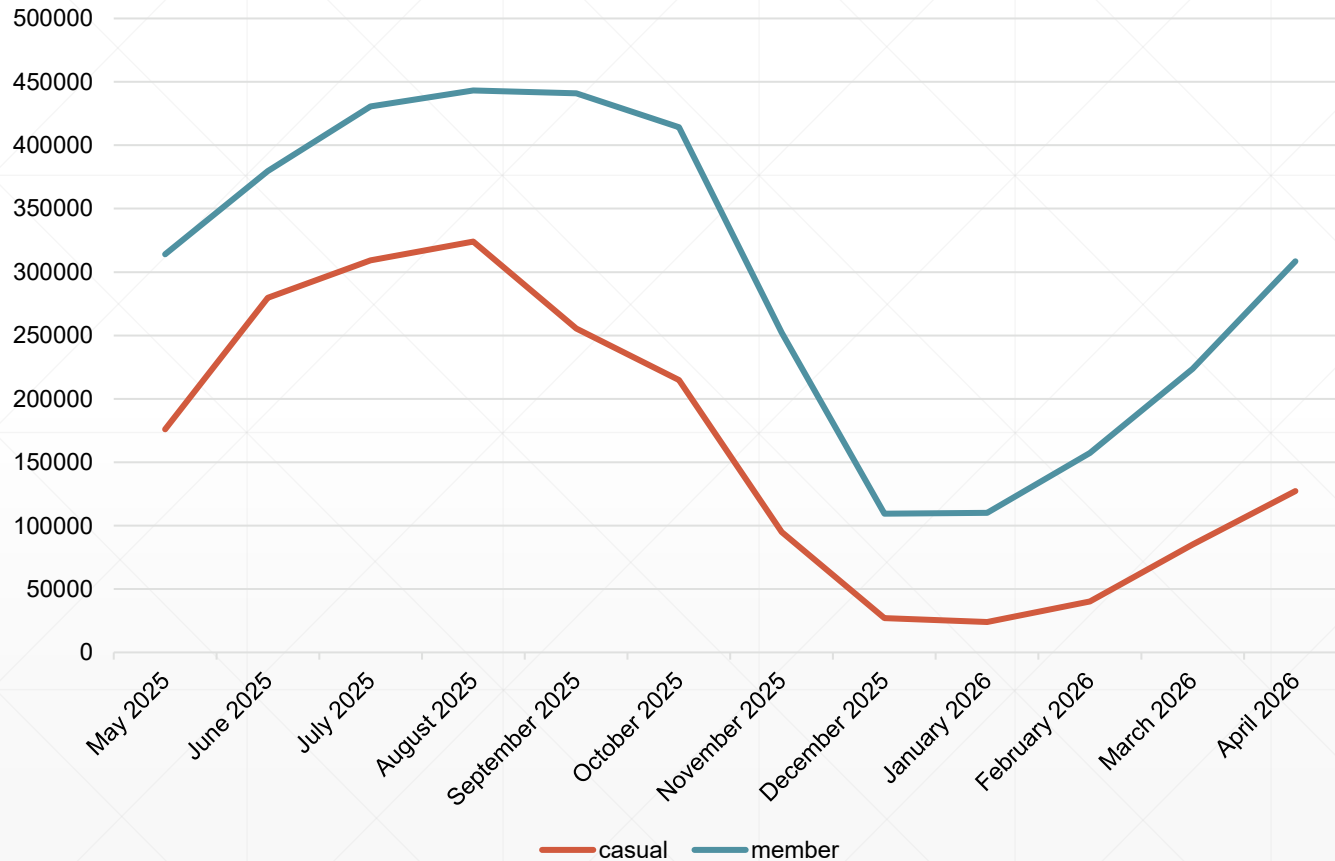
- Average ride duration of casual riders during weekdays ranges between 19 and 23 minutes, and increases during weekends
- Average ride duration of annual members is relatively stable throughout the week (12-14 minutes)
- This pattern suggests that casual riders primarily use Cyclistic bikes for leisure activities, while annual members use them for commuting

Number of rides by weekday



- Annual members complete a significantly higher number of trips
- Member rides peak during weekdays
- Casual rider activity increases during weekends

Number of rides by month



- Members consistently recorded more rides than casual riders throughout the year
- Ride activity peaks during the summer and early fall months (July-October).
- Ride activity for both groups declined significantly during winter months - weather conditions strongly influence bike usage

Conclusions

The analysis identified clear behavioral differences between annual members and casual riders:

- Casual riders use the service less frequently but spend significantly more time per ride than annual members
- Annual members complete a significantly higher number of trips
- Ride duration for casual riders increases considerably during weekends
- Average ride duration for annual members remains relatively stable throughout the week
- Weather conditions strongly influence bike usage

The analysis suggests that the greatest opportunity for membership growth lies among casual riders who regularly use Cyclistic bikes for recreational and weekend trips.

Recommendations

Based on the analysis, the following recommendations are proposed:

- Promote cost savings for frequent riders: develop marketing campaigns emphasizing the long-term financial benefits of annual membership compared to single rides.
 - Offer seasonal membership discounts: introduce discounted seasonal membership plans targeted at casual riders during high-activity months.
 - Launch weekend conversion campaigns: since casual rider activity peaks during weekends, targeted weekend promotions could encourage casual riders to become annual members.
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